

Wikipedia Contributions in the Wake of ChatGPT

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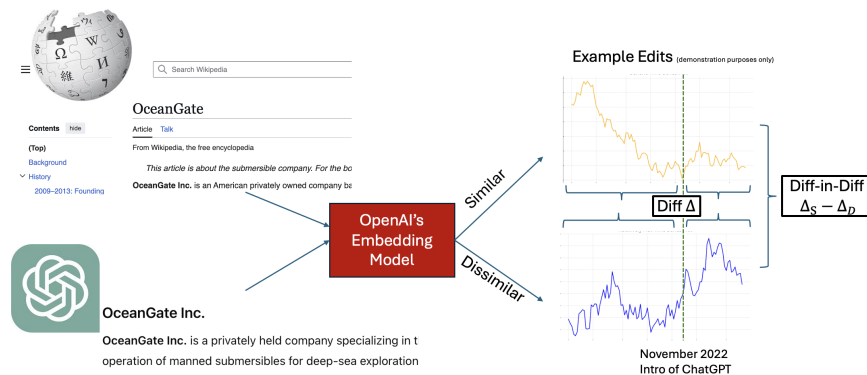


Figure 1: Our Diff-in-Diff Methodology. Measuring Wikipedia activity before and after the introduction of ChatGPT for Wikipedia articles that rank similarly and dissimilarly to the same information generated by ChatGPT.

Abstract

How has Wikipedia activity changed for articles with content similar to ChatGPT following its introduction? We estimate the impact using differences-in-differences models, with dissimilar Wikipedia articles as a baseline for comparison, to examine how changes in voluntary knowledge contributions and information-seeking behavior differ by article content. Our analysis reveals that newly created, popular articles whose content overlaps with ChatGPT 3.5 saw a greater decline in editing and viewership after the November 2022 launch of ChatGPT than dissimilar articles did. These findings indicate heterogeneous substitution effects, where users selectively engage less with existing platforms when AI provides comparable content. This points to potential uneven impacts on the future of human-driven online knowledge contributions.

*These authors have contributed equally.

CCS Concepts

• **Information systems** → Collaborative and social computing systems and tools; Information retrieval; Web mining; • **Human-centered computing** → Human-computer interaction (HCI); User studies; • **Computing methodologies** → Natural language processing; Language models.

Keywords

Wikipedia; ChatGPT; AI-generated content; human knowledge contributions; difference-in-differences; online collaboration; information seeking; substitution effects

ACM Reference Format:

Liang Lyu, James Siderius, Hannah Li, Daron Acemoglu, Daniel Huttenlocher, and Asuman Ozdaglar. 2025. Wikipedia Contributions in the Wake of ChatGPT. In *Companion Proceedings of the ACM Web Conference 2025 (WWW Companion '25)*, April 28-May 2, 2025, Sydney, NSW, Australia. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3701716.3715543>

1 Introduction

Generative AI tools like ChatGPT have fundamentally reshaped how information is disseminated and consumed online. Increasingly, users are turning to large language model (LLM) chatbots and AI-generated search summaries for information ([2]; [4]). This shift in behavior can create spillover effects on other platforms, as users substitute away from traditional crowdsourced knowledge sources such as Reddit ([1]) and Stack Overflow ([3]; [10]).



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ACM ISBN 979-8-4007-1331-6/2025/04
<https://doi.org/10.1145/3701716.3715543>

Importantly, these changes in user behavior can have downstream impacts not only on these traditional platforms, but also on future generative AI models. One concern is that if users substitute away from these platforms and viewership drops, then contributors may also be less inclined to create content and knowledge, which AI models are trained on. A decline in their quantity and quality can result in “model collapse” for future AI models [11].

In this study, we focus on Wikipedia, a knowledge sharing platform with unique concerns and learning opportunities regarding the potential negative effects of LLMs. Wikipedia’s scope extends far beyond technical, directed questions, instead offering an encyclopedic repository of historical, cultural and scientific topics upon which many other sources draw. The platform also hosts two distinct groups of users – viewers and editors – who could differ in their responses to LLMs and their downstream implications.

Several factors about Wikipedia necessitate our differences-in-differences (DiD) strategy, in contrast to the interrupted time series analysis that is often used in similar work on Reddit ([1]), Stack Overflow ([3]; [10]), and Wikipedia itself ([9]). In addition to having a broader scope of topics, Wikipedia allows for more diverse user incentives than analogous platforms: viewers exhibit both shallow and deep information needs [12], while contributors are driven by both intrinsic [14] and extrinsic motivations [13]. These factors may dampen the effects of ChatGPT on *some* users and articles. In fact, [9] analyze Wikipedia engagement in the aggregate, and do not identify significant drops in activity following the launch of ChatGPT. We hypothesize that their analysis do not fully capture the heterogeneity of Wikipedia, compared to similar platforms with more homogeneous contents and users.

The heart of our empirical strategy is that the impact of ChatGPT on Wikipedia should vary between different types of articles. We classify a subset of articles as “similar” to ChatGPT’s output, capturing the capability of the chatbot to learn and summarize relevant information in these articles. A non-overlapping subset of articles are classified as “dissimilar”, meaning that there is less possibility that ChatGPT will replace Wikipedia as the main provider of information for these articles. We then estimate DiD models with dissimilar articles as our “control group”, in order to estimate the causal effect of ChatGPT on viewership and edits.

More specifically, we define our two-year event study period centered at the launch of ChatGPT in November 2022, consider a pool of articles that had been popular before the end of this period and were created relatively recently, and gather their views and edits during the period. We pass their titles to GPT 3.5 and ask it to generate an encyclopedic article, and compute similarity scores between the ChatGPT outputs and the Wikipedia articles.

This empirical strategy relies on two assumptions. The first is the “stable unit treatment value assumption”, which states that our control group, dissimilar articles, is not impacted by ChatGPT. This assumption is motivated by the fact that ChatGPT is less capable of replicating the content of these articles. If the assumption is violated due to indirect effects (e.g., ChatGPT makes Wikipedia contributors more productive), our strategy will still reveal the differential additional effect on similar articles relative to dissimilar ones (but not the full impact on dissimilar articles). Our second assumption is the standard “parallel trends” assumption of DiD: absent ChatGPT,

similar and dissimilar articles would have evolved in similar fashion. Figure 2 below, which shows similar pre-ChatGPT evolution of these two groups, bolsters our confidence in this assumption.

Our analysis reveals that, following the launch of ChatGPT, articles with similar content experienced a significant decline in views relative to dissimilar articles. For edits, we find suggestive but not statistically significant evidence of fewer edits on similar articles compared to dissimilar ones. These findings suggest that substitution effects of ChatGPT may vary depending on its capabilities relative to the content of interest. It highlights the need for further research into the potential effects of LLMs on voluntary knowledge contributions across crowdsourced platforms, and mechanisms to explain such behavior for both viewers and content contributors.

2 Data and Methods

We perform a difference-in-differences (DiD) analysis of the growth in activities – views and edits in each month – of a pool of English Wikipedia pages from the 12-month period preceding ChatGPT’s launch to the 12-month period that follows it. Pages are classified as “similar” or “dissimilar” based on how much ChatGPT’s responses on the topic align with the actual Wikipedia pages. We analyze how the growths in activities differ for similar vs. dissimilar articles given fixed recency of articles, controlling for article length and the trend in overall growth of all articles.

2.1 Wikipedia Data

Time frame. We define our *event study* period E , the period during which activities are measured, as the 24 months from December 2021 to November 2023 (inclusive). It consists of the *pre-GPT* period up to ChatGPT’s launch date on November 30, 2022, and the *post-GPT* period starting from December 2022.

Articles. For each month up to November 2023,¹ we collect the 1000 pages with the highest number of views on English Wikipedia during that month. We obtain 20775 unique articles after data cleaning, denoted as set $\mathcal{A}$. The 2206 most recently created articles are used in subsequent analyses (since January 2020; see Section 2.3).

The choice of using the most popular articles per month is driven by the motivation to sample articles with substantial levels of activity, whose numbers of views and edits are representative and more robust to noise. Wikimedia APIs provide convenient access to top articles by monthly page views [6].

Measurements. For each article in $\mathcal{A}$, we measure its *activities*, in terms of views and edits, in each month of the event study period. An *observation* is defined as $(a, t, Views_{at}, Edits_{at})$, where $a \in \mathcal{A}$ is an article, $t \in E$ is a month in the event study period, and $Views_{at}$ and $Edits_{at}$ are a ’s number of views and edits in month t . We also collect the creation date and length of each article.

2.2 GPT Data and Similarity Score

For each article, we use GPT 3.5 Turbo [7] to generate an encyclopedic page given the title. This model, trained using data up to September 2021, was the primary model available to most ChatGPT users in 2023, and affected user decisions during the event study period more directly than later models.

We use the following prompt:

¹Starting with July 2015, the first month in which viewership data is available

You are an assistant whose task is to write an encyclopedic article for a given topic chosen by the user, similar to those found on Wikipedia. Generate an encyclopedic article in English with title "[title of actual Wikipedia page]".

We compute a *similarity score* for each article between its actual contents on Wikipedia and GPT 3.5’s imitation, as the cosine similarity of their text embeddings using OpenAI’s embedding model text-embedding-3-small. [8] The similarity score can be seen as a proxy for substitutability of the two options from the user’s point of view, and for GPT 3.5’s mastery of the topic.

2.3 Models

Recency. We parameterize the *recency* of articles in our sample by parameter T , and perform a series of regressions by varying T . For each T , an observation $(a, t, Views_{at}, Edits_{at})$ is called T -recent at time t if the article a was created no more than T months before t . We define $O(t, T)$ as the set of all T -recent observations recorded in month t , and $O(T) = \bigcup_{t \in E} O(t, T)$ the entire set of T -recency observations for some t in the event study period.

Varying the recency parameter T lets us analyze activities of articles of different ages. A small T means that we only look at articles created during or just before the event study period, and their activities soon after they were created. A larger T shifts the focus to older articles and their activities further away from creation.

Similarity classifications. We then compute similarity labels for each regression parameterized by T . Define $\mathcal{A}(T)$ as the set of all unique articles that are involved in observations $O(T)$. Each article $a \in \mathcal{A}(T)$ is then given a binary classification label, $s_T(a) \in \{0, 1\}$, such that $s_T(a) = 1$ iff its similarity score is above the median in $\mathcal{A}(T)$. These are considered “similar” articles, whereas those with $s_T(a) = 0$ are “dissimilar” articles. We apply 5% winsorization to views and edits (separately) of articles in $\mathcal{A}(T)$ to remove outliers.

Comparative time series. To analyze how activities of similar and dissimilar articles change over time, we fit the following regression model for a fixed T , with data as all observations in $O(t, T)$ for all months $t \in E$:

$$Views_{at} = \delta_l \text{length}(a) + \sum_{t'} \theta_{t'} \mathbf{1}_{CreationMonth(a)=t'} + \delta_t t + \epsilon_{at},$$

where δ_l and δ_t are linear trends for article length and activity time. An analogous model is defined for edits. We report the time series of mean residuals for each month of activity t aggregated over similar and dissimilar articles respectively, with bootstrapped standard errors. The results for $T = 6$ are presented in Section 3.1.

Difference-in-differences (DiD) analysis. We fit a regression model for each T to estimate the DiD using the same data as above:

$$Views_{at} = \gamma_1 s_T(a) + \gamma_2 \times \mathbf{1}_{t \geq \text{Dec } 2022} + \beta (s_T(a) \times \mathbf{1}_{t \geq \text{Dec } 2022}) + \delta_l \text{length}(a) + \sum_{t'} \theta_{t'} \mathbf{1}_{CreationMonth(a)=t'} + \delta_t t + \epsilon_{at}.$$

The coefficient of interest is β , which measures the effect of ChatGPT on similar articles. γ_1 , γ_2 and θ are fixed effects for similarity, activity occurring post-GPT, and creation month of the article, respectively. δ_l and δ_t are linear trends for article length and activity time. An analogous model is defined for edits.

To smooth the estimates for various T , we perform weighted least squares (WLS) where each observation (a, t) has weight $\alpha^{T-\text{Age}_{at}}$,

where Age_{at} is the age of article a at time t , for a chosen smoothing parameter α . The results are presented in Section 3.2.

3 Results

3.1 Comparative Time Series

Figure 2 shows a comparative time series of mean residual views and edits in each month during the event study period for articles that are at most $T = 6$ months old. For both views and edits, we see that similar articles exhibited little changes in activity from the pre-GPT to the post-GPT period (accounting for controls). Dissimilar articles, on the other hand, show an increase in edits after ChatGPT launched in November 2022 (Figure 2(b)). Views for dissimilar articles also show an increasing trend around the same time, with residuals even rising above those of similar articles (Figure 2(a)).

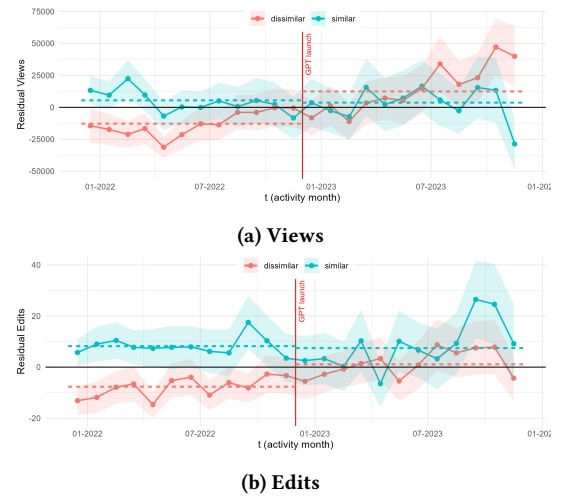


Figure 2: Mean residuals for each month t in which activity occurs, among all observations in $O(6)$ (at most 6 months old), with bootstrapped standard errors with 1000 samples. Dashed lines are mean residuals for similar (blue) and dissimilar (red) articles over the pre-GPT and post-GPT periods respectively. Red vertical line indicates date of ChatGPT launch.

3.2 Diff-in-Diff Regressions

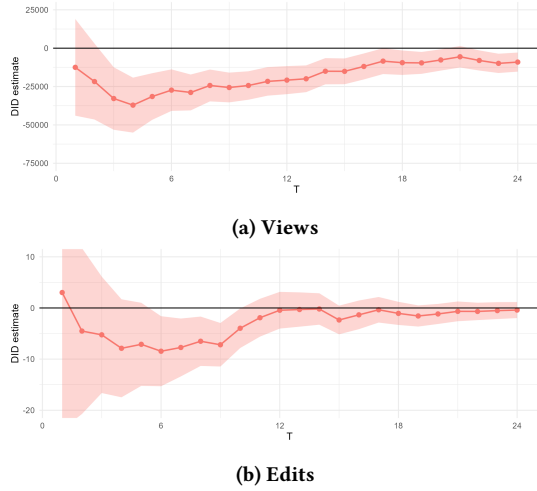
Figure 3 shows the estimated diff-in-diff coefficients for each recency parameter T from 1 to 24 months (x -axis), with smoothing factor $\alpha = 0.8$. Each point represents an individual regression: points to the left focus on newer articles and their views and edits in the first few months, while points to the right place greater weight on activities of older articles close to two years after creation. Table 1 shows the estimates for activities that occurred within at most $T = 6$ months since article creation, with $\alpha = 1$.

The diff-in-diff coefficients for Figure 3a (views) are negative and statistically significant for all article ages except $T = 1$, which implies that Wikipedia articles where ChatGPT provides a similar output experience a larger drop in views after the launch of ChatGPT. This effect is much less pronounced for edit behavior. The coefficients for Figure 3b (edits) are generally close to zero

Table 1: Diff-in-diff regression of views and edits with recency $T = 6$ (activities within 6 months of article creation), with $\alpha = 1$ (all articles given equal weight).

	Views	Edits
Diff-in-Diff Estimate	-29,013.340*** (6,903.964)	-6.877* (3.719)
Observations	6,060	6,060
R^2	0.069	0.140
Adjusted R^2	0.064	0.135

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

**Figure 3: Estimated DiD coefficients and 95% confidence intervals (HC1) for recency parameters $T \in \{1, \dots, 24\}$ on the x -axis, with smoothing factor $\alpha = 0.8$. Each point is from a regression using observations in $O(T)$. Observations when the article is closer to T months old given greater weight.**

and not statistically significant, particularly for articles older than 10 months. For newer articles less than 10 months old, the results are suggestive, but not necessarily significant, that similar articles receive fewer edits than dissimilar articles.

3.3 Interpretation

Both analyses show that similar articles that are created recently ($T \leq 10$) experience a greater drop in viewership and contributions post-GPT relative to dissimilar ones, with much stronger statistical significance for views. This shows that the impacts of ChatGPT are heterogeneous, and correlate with its substitutability over Wikipedia in providing information on relevant topics.

We also observe that the drop in views and especially edits of older articles are smaller in magnitude, evident from regressions with greater T . When $T > 10$, the effects on edits (if any) are no longer statistically significant, unlike views.

There are plausible mechanisms that could explain our findings, though an in-depth analysis of this question is beyond the scope of this paper. The majority of Wikipedia viewers have shallow

information needs such as a quick factual lookup [5, 12], a task that may be better suited to generative AI models. Contributors, on the other hand, are often driven by both intrinsic incentives [14] and extrinsic factors such as reciprocity [13]. Some editors, but not all, may thus become less motivated following declines in viewership.

4 Conclusion and Future Work

This work shows evidence of heterogeneous substitution patterns in Wikipedia viewership. The diff-in-diff analysis shows that articles that are similar to ChatGPT generated content receive fewer views after the launch of ChatGPT, compared to dissimilar articles. We also identify a suggestive, though inconclusive, trend in which these similar articles output receive fewer edits, particularly new articles.

The findings on edit activity could have important implications for AI models and warrant further study. A reduction in contributions on Wikipedia could degrade the quality of future training data. This phenomenon should be monitored over time, since contributor behavior may react more slowly than reader behavior. Other future directions include behavioral studies to explain the mechanisms behind the observed trends, and refinements in measuring substitutability of LLMs over Wikipedia as new models become more capable of learning and replicating Wikipedia content.

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